

11. Write a python program to create 2d array using NumPy

Title:

- Creating a 2D Array Using NumPy in Python

Objective:

- To create a 2D array using NumPy, which allows for efficient manipulation and mathematical operations on arrays.

Task:

- Import NumPy library.
- Create a 2D array with specific values or random values.
- Print the 2D array to verify its structure.

Code:

```
# Importing the numpy library
import numpy as np
# Creating a 2D array with specific values
array_2d = np.array([[1, 2, 3],
                    [4, 5, 6],
                    [7, 8, 9]])
# Alternatively, create a 2D array with random values (3x3 array)
# array_2d = np.random.randint(1, 10, (3, 3))
# Printing the 2D array
print("The 2D array is:")
print(array_2d)
```

12. Write a program to convert a python list to a NumPy array.

Title:

- Converting a Python List to a NumPy Array

Objective:

- To convert a Python list into a NumPy array, enabling more efficient mathematical operations and array manipulations.

Task:

- Import the NumPy library.
- Create a Python list.
- Convert the list to a NumPy array using the np.array() method.
- Print the resulting NumPy array.

Code:

```
# Importing the numpy library
import numpy as np
# Creating a Python list
python_list = [1, 2, 3, 4, 5]
# Converting the Python list to a NumPy array
numpy_array = np.array(python_list)
# Printing the NumPy array
print("The NumPy array is:")
print(numpy_array)
```

13. Write a program to create matrix of 4x5 from 11 to 30.

Title:

- Creating a 4x5 Matrix from 11 to 30 Using NumPy

Objective:

- To create a 4x5 matrix filled with numbers ranging from 11 to 30 using NumPy. This will allow you to explore matrix creation and indexing in NumPy.

Task:

- Import the NumPy library.
- Generate a sequence of numbers from 11 to 30.
- Reshape the sequence into a 4x5 matrix using np.reshape().
- Print the resulting 4x5 matrix.

Code:

```
# Importing the numpy library
import numpy as np

# Creating a sequence of numbers from 11 to 30
numbers = np.arange(11, 30 + 1)

# Reshaping the sequence into a 4x5 matrix
matrix_4x5 = numbers.reshape(4, 5)

# Printing the resulting 4x5 matrix
print("The 4x5 matrix is:")
print(matrix_4x5)
```

14. Write a program to create a data frame to store data of candidates who appeared in interviews. The dataframe columns are name, score, attempts, and qualify (Yes/No). Assign rowindex as C001, C002, and so on.

Title:

- Creating a DataFrame to Store Interview Data of Candidates

Objective:

- To create a pandas DataFrame to organize and store data of candidates who appeared in interviews. The DataFrame will contain candidate details such as name, score, number of attempts, and whether they qualify for the next round.

Task:

- Import the pandas library.
- Create a DataFrame to store candidate details.
- Define the columns as 'name', 'score', 'attempts', and 'qualify'.
- Assign custom row indices such as C001, C002, etc.
- Display the DataFrame to verify the results.

Code:

```
# Importing the pandas library
import pandas as pd

# Creating data for candidates
data = {
    'name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve'],
    'score': [85, 72, 90, 65, 78],
    'attempts': [1, 2, 1, 3, 2],
    'qualify': ['Yes', 'No', 'Yes', 'No', 'Yes']
}
```

```

}
# Assigning row indices as 'C001', 'C002', etc.
index_labels = ['C001', 'C002', 'C003', 'C004', 'C005']
# Creating the DataFrame
df = pd.DataFrame(data, index=index_labels)
# Printing the DataFrame
print("Data of candidates who appeared in interviews:")
print(df)

```

15. Consider the following data of a clothes store and plot the data on the line chart and customize the chart as you wish:

Month	Jeans	T-Shirts	Shirts
March	1500	4400	6500
April	3500	4500	5000
May	6500	5500	5800
June	6700	6000	6300
July	6000	5600	6200
August	6800	6300	4500

Title:

- Plotting Sales Data of Clothes Store on a Line Chart

Objective:

- To represent the sales data of various clothing items (Jeans, T-Shirts, and Shirts) over several months on a line chart, and customize the chart with a title, axis labels, and legends to make it visually appealing.

Task:

- Define the data for each clothing item (Jeans, T-Shirts, and Shirts) for the months from March to August.
- Plot the sales data on a line chart.
- Customize the chart with appropriate labels for the axes, a title, and a legend.

Code:

```

# Importing necessary libraries
import matplotlib.pyplot as plt
# Data for each item in different months
months = ['March', 'April', 'May', 'June', 'July', 'August']
jeans_sales = [1500, 3500, 4500, 5500, 6000, 6800]
t_shirts_sales = [4400, 5000, 6500, 6700, 6000, 5600]
shirts_sales = [6500, 5800, 5000, 6300, 6200, 4500]
# Plotting the data
plt.plot(months, jeans_sales, label='Jeans', marker='o', linestyle='-', color='blue')
plt.plot(months, t_shirts_sales, label='T-Shirts', marker='s', linestyle='--', color='green')
plt.plot(months, shirts_sales, label='Shirts', marker='^', linestyle='-.', color='red')
# Customizing the chart
plt.title('Sales Data of Clothing Items (March - August)')
plt.xlabel('Month')
plt.ylabel('Sales (in units)')
plt.grid(True)
plt.legend()
# Display the plot
plt.show()

```